

Frame Relay to ATM interworking

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Frame Relay is a well-established technology that has attracted a large number of customers world-wide. It is typically used to interconnect large, multi-site company local area networks (LANs). These data connections are usually at low to medium speeds. Asynchronous transfer mode (ATM) is a newer, broadband technology that is capable of very high speed LAN interconnection. In addition, it is arguably an ideal transport medium for legacy protocols such as Frame Relay.

This paper describes the data transport features of each technology and discusses how ATM may be used to interconnect existing Frame Relay networks, and how an ATM user may exchange data with a Frame Relay user.

1. Introduction

The history of Frame Relay takes us back to 1976, when X.25 was created. X.25 specifies the interface between a terminal and a packet switched data network. Packet switching is well suited to data transmission because of its ability to statistically multiplex packets intended for different physical destinations over a single physical link, thus eliminating the problem of needing $\sim N^2$ number of access lines for interconnection of N sites. An additional benefit of sending a data stream in packets is that it is easy to detect errors at the destination by means of checksums, and to overcome errors using packet re-transmission. In the 1970s the transmission paths used were analogue and suffered high error rates, which made error correction on a link-by-link basis desirable (as it is inefficient to propagate errored packets through the network). X.25 takes these considerations into account, but the complex protocol processing used tends to limit the maximum data rates available and to incur relatively large end-to-end delays.

Frame Relay is a 'lightweight' protocol that is designed to give a higher performance over digital networks, which have lower error rates. Processing delay in the switch is reduced by the simplicity of the protocol. There are a large number of Frame Relay accesses deployed world-wide, which are primarily used for LAN interconnect. Frame Relay is designed to cope with 'bursty' traffic, suiting the profile of inter-LAN traffic.

ATM has been chosen by the ITU-T to be the technology used to support broadband integrated services digital networks (B-ISDNs). The key features of ATM are bandwidth flexibility, the support of voice, video and data integration, and quality of service guarantees for different classes of service.

This paper compares the technologies used by both Frame Relay (FR) and ATM and then discusses how the two protocols may communicate with each other. This may be done by two methods — either by transporting FR over ATM transparently, which is termed network interworking, or by converting the FR traffic to ATM traffic at the FR/ATM boundary, which is termed service interworking.

2. The technologies

2.1 Frame Relay

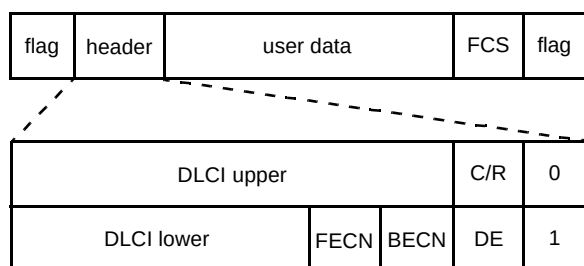
Frame Relay has its origin in the X.25 packet switching protocol. X.25 is a 'heavyweight' protocol suite that consists of specifications for the first three layers of the OSI 7-layer model for a terminal to access a packet-switched data network. The layer 2 protocol used in X.25 is LAPB (link access protocol B), a subset of HDLC (high level data link control), which provides error-free transport between packet switching exchanges on a link-by-link basis. The X.25 packet layer protocol (PLP) at layer 3 is used to provide error and flow control.

Over recent years, improvements in transmission networks leading to virtually error-free layer 1 data transmission has shifted the emphasis to 'lightweight' lower layer protocol suites. Applying this philosophy to X.25, the layer 3 protocol is eliminated and the link layer is reduced to a minimal 'core' LAPB protocol, effectively shifting the added functionality of the X.25 PLP and LAPB into the higher layers of the ISO protocol stack at the end-points of the network path. The remaining layer 2 protocol is known as 'Frame Relay' and supports:

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- frame delineation (using HDLC flags),
- zero bit insertion,
- CRC checking,
- multiplexing and de-multiplexing of many virtual circuits on to a single physical link,
- indications for forward and backward congestion,
- an indication for low priority frames.

The format of a frame is shown in Fig 1.



BECN backward explicit congestion
 C/R command/response
 DE discard eligibility
 DLCI data link connection identifier
 FCS frame check sequence
 FECN forward explicit congestion

Fig 1 Structure of an FR frame.

Frame Relay is commonly used to transport IP traffic between LANs. IP packets up to a maximum size of several thousand bytes are encapsulated into frames which are then carried, typically by a leased line, into the local FR point of presence. Access rates of 64 kbit/s to 2 Mbit/s are common, although support for E3/T3 rates (34/45 Mbit/s) by vendors is imminent. Individual frames are then switched, based on the link-layer address (DLCI). The DLCI is usually 2 octets in length, though up to 4 octets may be supported. Even so, this is not sufficient to form a global addressing scheme and so is of local significance only. A Frame Relay user subscribes to a committed information rate (CIR) from the network operator which is the rate at which the network agrees to transfer data under normal conditions. The user may send traffic at a higher rate than this, with the excess traffic being marked as low priority (DE bit set to 1) by the network. This excess traffic is discarded first under network congestion conditions.

As with X.25, early Frame Relay network devices were based on store-and-forward packet-switching techniques. Using this method an entire frame would be buffered until the end checksum was received and then the frame would be sent over the next network trunk link. However, buffering delays in Frame Relay networks can be reduced by segmenting the frames into fixed length cells and forwarding the cells immediately, reassembling the cells on

arrival at the destination switch. This is effectively network interworking over ATM (see section 4).

Although Frame Relay is designed for non-real-time packetised data transmission, there has been much interest in using Frame Relay to carry voice. As voice traffic is highly sensitive to delay, it is necessary to send bursts of voice samples using short length frames. One problem with this method is that the short voice frames could be queued behind the much longer data frames at various points in the network. Two potential solutions to this are either to constrain the size of all frames on the network to a short, fixed length, or to implement a mechanism where short voice frames interrupt long data frames on the access line. In both cases voice frames have a higher priority path throughout the network. Both of these mechanisms are currently non-standard, but the concepts of fixed length frames and priority queuing are used in a standardised way in ATM to effectively support all traffic types, including voice.

2.2 ATM

ATM is a flexible protocol which can support both constant bit rate (CBR) or variable bit rate (VBR) connections. It supports a number of quality of service (QoS) classes in terms of delay, loss and individual connection bandwidth. The QoS provided to an end user is guaranteed by means of the underlying connection-oriented nature of ATM. This requires that a virtual connection (VC) is set up before user information can be transferred between end users. A VC can be set up permanently (a PVC), by the network administrator, or on a per-call basis (a switched VC or SVC [1]) by the user signalling to the network.

ATM uses short, fixed length cells of 53 bytes in which 48 bytes are used for the payload and 5 bytes are used as the cell header. The relatively low complexity of the switching protocol permits a fast hardware implementation. The format of the cell header for the UNI is shown in Fig 2. The GFC field may be used to support a flow-control mechanism. Connections are identified using a virtual path identifier and virtual circuit identifier (VPI/VCI) which have local significance, as with the FR DLCIs. Low priority cells are marked (tagged) with the cell loss priority (CLP) bit, and forward congestion is indicated by setting the

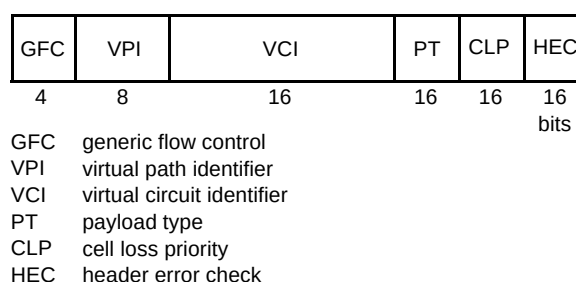


Fig 2 Structure of an ATM cell.

explicit forward congestion notification (EFCI) bit. A payload type (PT) field indicates whether the cell payload contains user information or management information.

ATM adaptation layers (AALs) are used to interface higher layer protocols to the ATM layer. AAL-5 is used to support FR services. Separate queuing buffers for each service class are used on the input and/or output of the switching matrix.

3. Standards activities

In 1993 the ITU-T produced Recommendation I.555 [2] which describes the various scenarios involved when interworking Frame Relay with B-ISDN (i.e. ATM).

The Frame Relay Forum (FRF) is an industry forum that issues Implementation Agreements which can be used by vendors to generate products that successfully interwork. FRF.5 [3] and FRF.8 [4] build on the scenarios identified by I.555 for FR-to-ATM network and service interworking respectively. There is a draft agreement specifying SVCs using Frame Relay to ATM service interworking [5].

The ATM Forum has produced a traffic management specification for the Broadband Inter-Carrier Interface (B-ICI), and Appendix A [6] of this provides guidelines on the conversion of traffic parameters from FR to ATM.

The IETF has produced a specification for the encapsulation of multiple upper layer protocols by FR (RFC 1490 [7]) and by ATM (RFC 1483 [8]).

4. FR to ATM interworking

There are two types of Frame Relay to ATM interworking defined — network interworking and service interworking.

4.1 Network interworking

This is used to connect two Frame Relay users over ATM (Fig 3). The Frame Relay frames are encapsulated in the ATM transport protocol. Network interworking occurs at the ingress and egress points of the ATM network. It should be noted that it would also be possible to perform network interworking if one of the Frame Relay users were connected by an ATM access line, but emulating the Frame Relay protocol in the ATM CPE. However, if this were required, it would be simpler to use service interworking, as described below in section 4.2.

One means of interconnecting FR switches is to use leased lines. However, there are advantages of using ATM to interconnect FR switches instead of leased lines:

- ATM offers a finer bandwidth granularity, e.g. connections could be specified to the nearest 1 kbit/s, whereas leased lines are commonly available at 2, 8, 34 or 140 Mbit/s data rates,
- the ATM connection data rate can be quickly increased or decreased,
- ATM can be used as a core technology that supports a wide range of services (e.g. FR, SMDS, voice, video) and the cost of the core network can be distributed among all the services,
- connections between FR switches may be multiplexed on to a single ATM access link rather than using a number of leased lines.

Network interworking also provides an efficient means to interconnect Frame Relay switches by permitting the statistical gain that arises from the aggregation of packetised traffic. Thus, the bursty Frame Relay application traffic may be carried more cost effectively using a VBR

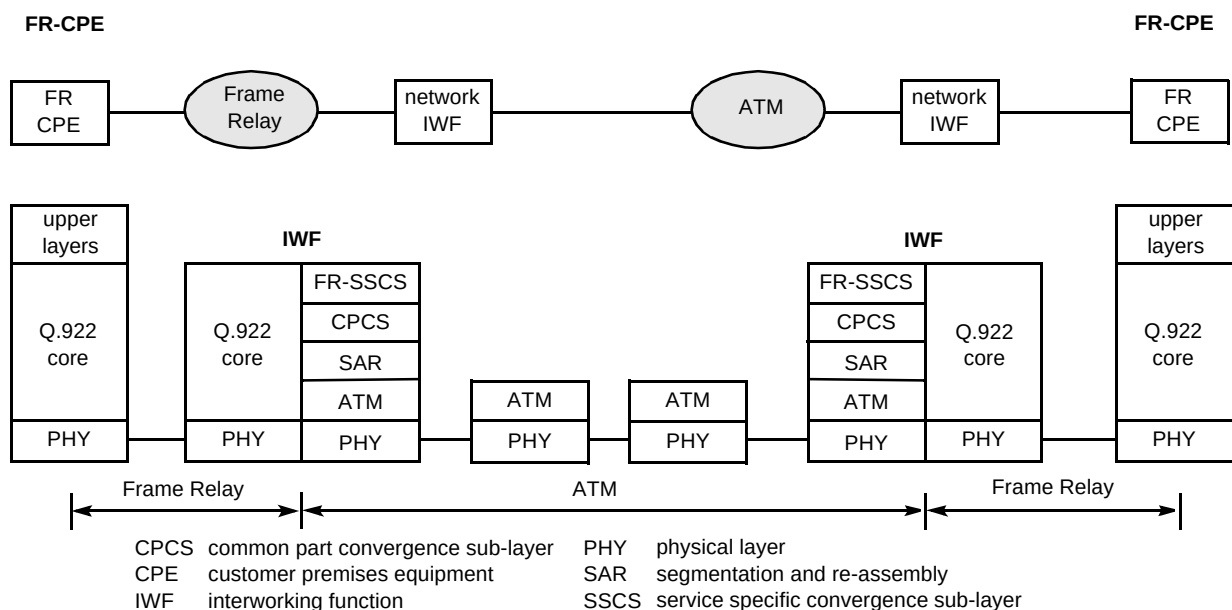


Fig 3 Network interworking and associated protocol stacks.

ATM service class which is designed to support this type of traffic profile, e.g. tagged VBR.

4.2 Service interworking

This is used to connect a Frame Relay user to an ATM user and vice versa (Fig 4). A single service interworking function performs a mapping between the Frame Relay protocol and the ATM protocol.

There are a large number of users currently connected to Frame Relay services. Often, these users are large companies with multiple sites which may be within a single country or may be world-wide. In the future these users may wish to upgrade their network by changing the access protocol of some sites to ATM, or by introducing new ATM sites. This may be particularly important for central hub sites requiring the much higher bandwidths that ATM can offer, or for sites wishing to use ATM for integrated access (e.g. voice and data). Therefore, service interworking will be required as some sites will use Frame Relay and some sites will use ATM but they will both need to share data.

5. Network interconnection

To facilitate service interworking, an FR network may be connected to an ATM network using either the FR or ATM protocol, the former requiring the ATM switch and the latter requiring the FR switch to support service interworking. To enable both to be used, a number of additional factors need to be considered.

5.1 Link speed

Interconnection using Frame Relay is currently possible using comparatively low speed links of the order of 2 Mbit/s. ATM permits much higher speeds to be used, including E3 (34 Mbit/s) and STM-1 (155 Mbit/s).

5.2 Link resilience

With service interworking it is important to consider the overall resilience of an FR-to-ATM path. Generally FR or ATM switches from a given manufacturer will include proprietary routing protocols to ensure that a PVC path-failure will be re-routed automatically. However, this automatic re-routing is generally not possible across multi-vendor networks. A solution to this problem may be realised when FR-SVC to ATM-SVC interworking becomes possible (an example of multiple vendor interworking for all-ATM networks is given in Clarke et al [1], and for all-FR networks is given in the Frame Relay Forum agreement FRF.10 [9]). Until these become available on the switches in use, the failure of a path, which can be detected by loss of signal at the physical layer and by loss of OAM cell flows or loss of frames at layer 2, can be dealt with using operational support systems (OSS) processes. These occurrences are expected to be rare because of the transmission resilience which may be provided, for example, by using SDH rings [10] at the physical layer.

5.3 Network functionality

If ATM is used as the network interconnection protocol, then, in addition to supporting user applications that utilise the standards-based Frame Relay traffic parameters, the

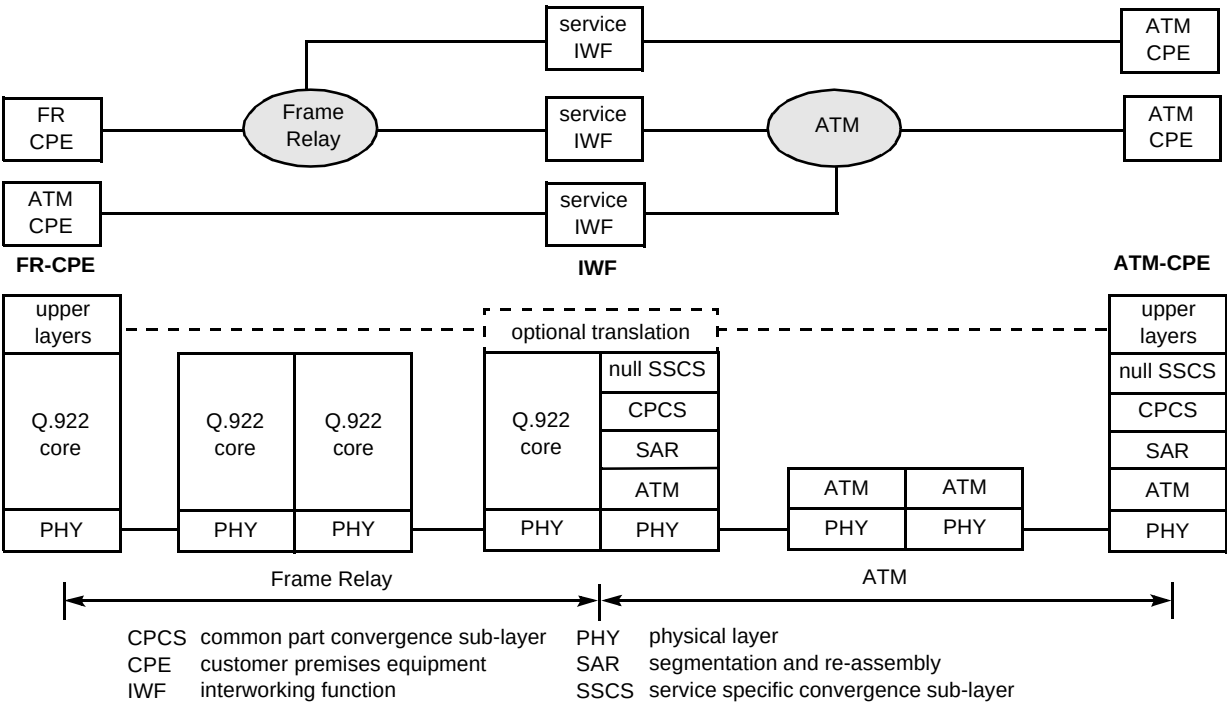


Fig 4 Service interworking and associated protocol stacks.

interface and the ATM network will be able to support any proprietary extensions that vendors of FR switches may include (e.g. voice over Frame Relay). If FR is used as the network interconnection protocol, the FR standards-based interface on the ATM switch will not be able to support any proprietary extensions used on the FR switches.

5.4 Link cost

The use of higher speed ATM links for interconnection means that fewer physical interfaces would be required than if FR were used. This is important, but the overall cost depends partly on the cost of ATM interface cards compared with Frame Relay cards for both networks.

In short, it is preferable to interconnect the networks with ATM which is able to support multiple service classes at high speeds.

6. Parameter mappings

6.1 Network interworking

In this case there are two interworking functions in any path through the networks and ATM can be considered to be a transport protocol that carries the FR frames in between the IWFs. The FR frames are sent over ATM virtually transparently, but because both FR and ATM are layer 2 protocols in the OSI seven-layer reference model, there needs to be some interaction between the protocols. Upper layer protocols (e.g. IP) are carried transparently. The IWF (interworking function) includes the following functions (also refer to FRF.5 [3] and Table 1).

Table 1 Network interworking parameter mappings.

Mapping function	FR-to-ATM direction	ATM-to-FR direction
Frame formatting and delimiting	FR frame mapped into an FR-SSCS PDU: - FR frame flags, inserted zero bits and CRC-16 are removed, - support of 2-octet address field compulsory, support of 3- and 4-octet address fields is for further study.	AAL5 messages used to identify frame boundaries. Zero bits, CRC and flags inserted. Fields of AAL5 PDU mapped into the fields of FR frame.
Discard eligibility and cell-loss priority	Two modes of operation: - mode 1 (mandatory) — DE field is copied unchanged to the DE field of the FR-SSCS PDU, and is mapped to the ATM CLP field of every cell that comprises a specific frame, - mode 2 (mandatory) — DE field is copied unchanged to the DE field of the FR-SSCS PDU, and the ATM CLP of every cell that comprises a specific frame is set to a constant value (0 or 1) configured at service subscription time.	Two modes of operation: - mode 1 (mandatory) — if one or more cells of a frame has its CLP field set, or if the DE field of the FR-SSCS is set, the IWF shall set the DE field, - mode 2 (optional) — the FR-SSCS DE field is copied unchanged to the DL-CORE DE field. This is independent of the value of the ATM CLP bit.
Congestion indication — forward	Frame level FECN is not mapped to cell level EFCI (which is not set). The DL-CORE FECN is mapped unchanged to the FR-SSCS FECN.	Cell level EFCI is mapped to frame level FECN, which is set if the last cell received has EFCI set, or if the FECN field in the FR-SSCS PDU is set.
Congestion indication — backward	The BECN field of the FR-SSCS PDU is set if either the DL-CORE BECN field is set or the EFCI was set in the last ATM cell of the last frame received in the ATM-to-FR direction (a mechanism such as a timer may be needed to exit certain states, depending on the traffic activity).	The BECN field has no equivalent ATM protocol field. The FR-SSCS PDU BECN is copied to the DL-CORE BECN.
DLCI field (connection multiplexing)	One-to-one — each DLCI is mapped to a single ATM BCC. Multiplexing is performed at the ATM layer using VPI/VCI combinations. The DLCI (in range 16-991) is agreed upon by the two end systems or the default value of 1022 is used. Many-to-one — multiple DLCI values are carried by one ATM VCC. This is only used for FR PVCs that terminate on the same ATM-based end systems.	One-to-one — multiplexing done using VCCs. Many-to-one — multiplexing done using DLCIs.

- Variable length PDU formatting and delimiting

Referring to Fig 3, frames received by the IWF from the FR user are converted into a Frame Relay Service Specific Convergence Sub-layer Protocol Data Unit (FR-SSCS PDU) format which is essentially the same as a Q.922 frame but without the CRC-16, HDLC flags and zero bit insertion. The FR-SSCS PDU is passed to ATM adaptation layer 5 (AAL-5) before the segmentation and reassembly layer (SAR) converts the variable length PDU into fixed length cells of 53 bytes.

- Error detection

Errors between the two IWFs are detected using the AAL-5 32-bit CRC. The IWFs also maintain a log of certain error conditions, such as FR-SSCS PDU length errors, CRC errors and reassembly time-outs.

- Connection multiplexing

Two methods may be used. In the 'one-to-one' method each FR DLCI address is mapped to a separate ATM VPI/VCI address. In the 'many-to-one' method multiple FR connections are carried using a single VPI/VCI value. The one-to-one method allows FR PVCs to terminate on different ATM end-points, but the many-to-one method is simpler to set up for FR PVCs that terminate on the same ATM end-point, as only a single VPI/VCI is needed.

- Loss priority indication

FR frames use the discard eligibility (DE) bit to indicate low priority frames, while ATM cells use the cell loss priority (CLP) bit to indicate low priority cells. In the FR-to-ATM direction, the network provider can choose between mapping the DE value directly into the CLP field of all cells generated by a frame, or to set the CLP bit to a constant value which is used for all cells generated by the segmentation of all frames (although the CLP bit may still be modified by the ATM policer). In the ATM-to-FR direction, the network provider can choose to either set the DE bit to 'low priority', if any cells comprising a re-assembled frame have their CLP bit set to 'low priority', or can carry the DE bit transparently through the network.

- Congestion indication

Traffic congestion is indicated in FR networks using the FECN bit in the forward direction and the BECN bit in the backwards direction. ATM cells use the EFCI bit to indicate congestion in the forward direction and have no means of indicating congestion in the backwards direction. However, in the FR-to-ATM direction the FECN is not mapped to EFCI, which is always set to 'no congestion'. In the ATM-to-FR direction the EFCI is mapped to the FECN.

In the ATM-to-FR direction, the BECN field of the transported FR-SSCS PDU is copied unchanged to the BECN field of the reassembled FR frame. In the FR-to-ATM direction the BECN field of the FR-SSCS is set to 'congestion experienced' if either the frame has BECN set or if EFCI was set to 'congestion experienced' on the last ATM cell received in the opposite direction on the same virtual circuit.

- PVC status management

The management of PVCs within an FR network is described by Q.933 Annex A [11]. The management of PVCs within an ATM network is described by I.610 [12]. FRF.5 states that the management of the ATM layer and the FR-SSCS layer can operate independently, although the ATM VPI/VCI status indication obtained by OAM F5 information flows is used by the FR-SSCS layer when determining the status of FR PVCs. DLCI 0 is used to exchange management information between the IWFs, using bi-directional procedures.

6.2 Service interworking

In this case both FR and ATM can be considered 'peer' protocols which, when they meet at the IWF, are entirely mapped from one to the other. The IWF may be a part of the ATM network, part of the FR network, or separate from the networks. The ATM user has no knowledge that the other user has FR access, and vice versa — upper layer protocols are carried transparently. There are two modes of working.

- Transparent mode

This is where FR user data and AAL-5 user data are passed without modification. This is useful in environments where both FR and ATM users are not using protocol encapsulation at layer 2, or where the type of encapsulation used is known only to the end systems.

- Translation mode

This can be used for carrying multiple upper layer user protocols where the FR encapsulation conforms to RFC.1490 and the ATM encapsulation conforms to RFC.1483. The IWF then maps between the two upper layer protocol identifier formats, without examining the upper layer protocols. Illustrations of how an IP packet is encapsulated using both RFC.1490 and RFC.1483 appear in Figs 5 and 6.

The functions carried out by the IWF have some differences to those described above under network interworking (also refer to FRF.8 [4] and Table 2).

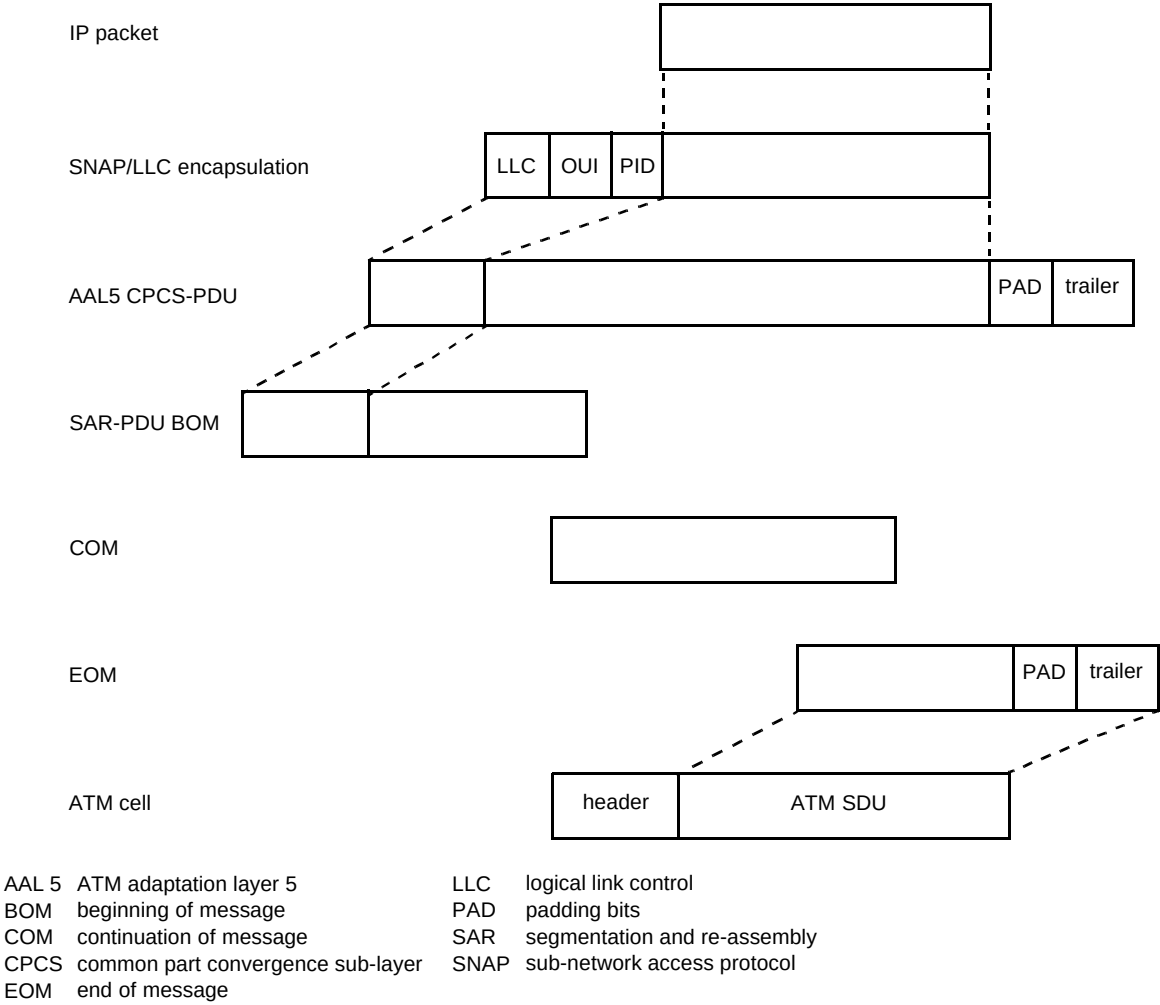


Fig 5 An IP packet carried over ATM.

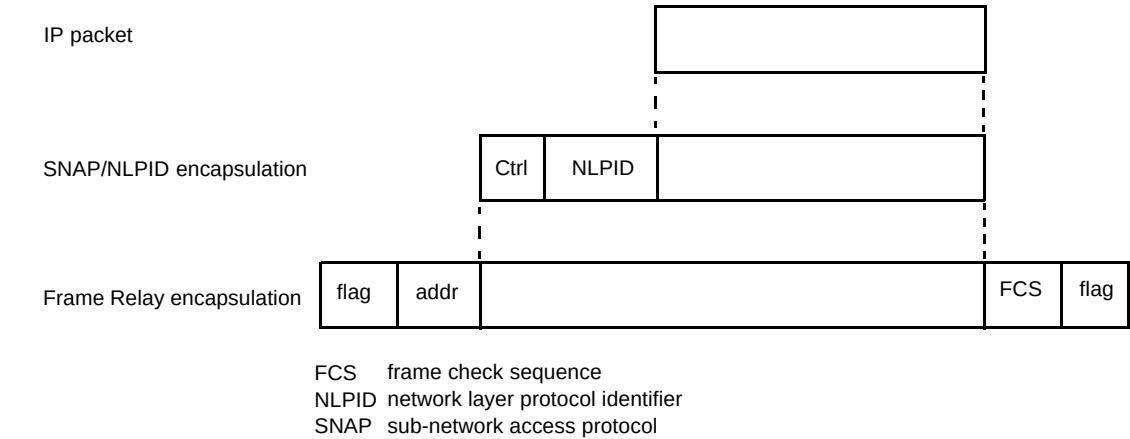


Fig 6 An IP packet carried over Frame Relay.

Table 2 Service interworking mappings.

Mapping function	FR-to-ATM direction	ATM-to-FR direction
Frame formatting and delimiting	FR frame mapped into an AAL5 PDU FR frame flags, inserted zero bits and CRC are removed. Q.922 core header removed and some of the header fields mapped into the ATM header fields.	AAL5 messages used to identify frame boundaries. Zero bits, CRC and flags inserted. Fields of AAL5 PDU mapped into the fields of FR frame.
Discard eligibility and cell-loss priority	Two modes of operation: - mode 1 (mandatory) — DE field mapped to ATM CLP field of every cell that comprises a specific frame, - mode 2 (optional) — the ATM CLP of every cell that comprises a specific frame is set to a constant value (0 or 1) configured at service subscription time.	Two modes of operation: - mode 1 (mandatory) — if one or more cells of a frame has its CLP field set, the IWF shall set the DE field, - mode 2 (optional) — the DE field shall be set to a constant value (0 or 1), configured at service subscription time.
Congestion indication — forward	Two modes of operation (both shall be supported): - mode 1 — the FECN field mapped to the ATM EFCI field of every cell that comprises a specific frame, - mode 2 — the FECN field is not mapped, the ATM EFCI field is always set to ‘congestion not experienced’. Note: for both modes, if there is congestion in the forward direction in the ATM layer within the IWF, the IWF can set the EFCI field to ‘congestion experienced’.	If the EFCI field in the last cell of a segmented frame is set to ‘congestion experienced’ then FECN is set to ‘congestion experienced’.
Congestion indication — backward	The BECN field has no equivalent protocol field. The IWF shall ignore the BECN.	The BECN field has no equivalent protocol field. The IWN shall set the BECN bit to zero.
Command/response field	The C/R field of the frame is mapped to the CPCS-UU least significant bit.	The CPCS-UU least significant bit is mapped to the C/R field.
DLCI field	The association between the FR DLCI and the ATM VPI/VCI is made when the PVC is provisioned.	The association between the FR DLCI and the ATM VPI/VCI is made when the PVC is provisioned.

- Variable length PDU formatting and delimiting

This is the same as for network interworking.

- Error detection

The FR CRC-16 is used to detect errors across the FR link segment, while the ATM AAL-5 CRC-32 is used to detect errors across the ATM link segment.

- Connection multiplexing

The ‘one-to-one’ method is used. Each FR DLCI address is mapped to a separate ATM VPC/VCI address.

- Loss priority indication

The DE and CLP bits can either be mapped or set to a constant value.

- Congestion indication

In the ATM-to-FR direction, the BECN bit is always set to zero, with the FECN and DE bits being either mapped from the EFCI and CLP bits respectively, or set to a constant value. In the FR-to-ATM direction BECN is ignored as it has no equivalent field in ATM; the EFCI is either mapped from FECN or set to a constant ‘congestion not experienced’ and CLP is either mapped from DE or set to a constant value.

- PVC status management

On the FR side the management procedures are as defined in Q.933 Annex A with bi-directional procedures being used. Optionally, asynchronous status messages may be sent. On the ATM side the procedures are as defined in ATM Forum UNI 3.1 [13] and B-ICI 1.1 [14].

- Other mappings

In the ATM-to-FR direction the AAL-5 CPCS user-to-user least significant bit is mapped to the Q.922 command/response field. In the FR-to-ATM direction the Q.922 command/response field is mapped to the user-to-user least significant bit.

6.3 Traffic parameter mapping

Frame Relay and the tagged VBR service on ATM are quite similar services with respect to traffic contract parameters. Both provide a guaranteed information rate to the user while allowing an additional information rate, provided there is spare network capacity.

Frame Relay traffic parameters are defined in terms of a committed information rate (CIR), an access rate (AR), and a committed burst size (B_c). The AR is the line rate, and is the speed at which individual frames are sent. Idle patterns are sent in between the frames. The CIR is the maximum rate at which frames may be sent without causing any frames to be tagged by the network as eligible for discard. B_c is the maximum length of contiguous data that may be sent at line rate before the frame containing the data is tagged as eligible for discard.

ATM parameters, for a tagged VBR service suitable for supporting this FR traffic, consist of a sustained cell rate (SCR), a peak cell rate (PCR), and a maximum burst size (MBS). The PCR is the maximum rate at which cells may be transmitted from CPE without being discarded on ingress to the ATM network by the policer. The SCR is the maximum long-term rate at which cells may be sent without tagging by the ATM policer. The MBS is the maximum number of contiguous cells that may be sent at a rate of PCR without tagging of any cells.

Therefore in the FR-to-ATM direction, assuming that the FR switch does not shape the user traffic, the AR can be mapped to the PCR, the CIR can be mapped to the SCR and the B_c can be mapped to the MBS. Assuming that a dual 'leaky bucket' policer is used on the ATM switches then the mappings defined in the ATM-F B-ICI specification [6] apply. Simplified here, these are:

$$\begin{aligned}\text{PCR} &\propto \text{AR} \\ \text{SCR} &\propto \text{CIR}\end{aligned}$$

with proportionality (or 'overhead') factors which allow for frame-to-cell conversion. The MBS is related to the B_c and the ratio of the CIR to the AR. The overhead factors are dependent on the number of information octets in the frame, and will be greater for small frames because the fraction of ATM bandwidth used by AAL-5 padding octets will be greater. In a typical example, using the TCP/IP protocol over Ethernet access, the frame size could range from 41

bytes for TCP acknowledgement packets to around 1500 for file transfers over Ethernet or 4000 bytes for 4 Mbit/s token ring. These figures correspond to a range of 2 to 84 cells per FR frame.

The VBR tagged service class may be enhanced for frame-based traffic by the inclusion of a frame-based policer for ingress traffic to the ATM network, coupled with a frame-based discard mechanism if congestion is encountered within the network, such as EPD (early packet discard). Both the policer and EPD will prevent cells which cannot reassemble into whole frames from traversing the ATM network. This situation could arise when other cells belonging to the same frame are discarded due to exceeding the ATM traffic contract or through congestion. The proposed GFR (guaranteed frame rate) service class, currently being standardised, is also expected to include these features.

7. Conclusions

This paper has described the relevant data transport features of ATM and Frame Relay and how these networks may be connected together.

Network interworking may be used to connect together Frame Relay switches using a high-speed, resilient and scalable ATM core network which offers bandwidth flexibility both in terms of an appropriate service class (VBR tagged) that matches the bursty data typically used by end applications, and by the ability to quickly increase or decrease bandwidth by agreement with the network operator. The maintenance costs can be distributed between multiple services. Overall, the cost of the network infrastructure is expected to be cheaper than using the alternative of leased line interconnection.

Service interworking may be used to connect together ATM service users with Frame Relay service users. Layer 3 information may be passed transparently either at the bit level, or in a multiprotocol environment by conversion between the different encapsulation formats. In all cases each service user is unaware that the other is using a different access protocol. This offers existing users of Frame Relay a convenient migration path to ATM and is expected to be implemented by companies with multiple branch offices that require higher bandwidth or multiservice capability at a central site. Branch offices can continue to use Frame Relay services with existing hardware (e.g. routers with Frame Relay interfaces) while central sites can upgrade their interfaces from Frame Relay to ATM.

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- 14 ATM Forum: 'B-ICI Specification, V1.1', (September 1994).



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